

Losing the War of the Future

*How New Technologies Threaten
America's Military Advantage*

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In its recent campaign against Iran, the United States dominated the skies using its traditional airpower. The U.S. military pounded Iranian targets, conducting over 13,000 strikes. That prowess and devastating

firepower did not stop Iran from hitting back. Over the course of the 39-day conflict that began on February 28 and stopped on April 8, Iran launched over 2,200 missiles and 4,400 drones against countries in the region. At least eight U.S. aircraft were destroyed or damaged by Iranian attacks. Multiple U.S. radars were hit, and seven U.S. service members were killed. And at the time of this writing, the Iranian regime remains in place and maintains a stranglehold over the Strait of Hormuz. The United States has not achieved its objectives in the war, even though it is by every metric far more powerful than Iran.

The technological dominance that the U.S. military has long counted on to give it an advantage over competitors is waning. Unlike in past eras, when the [United States](#) maintained major leads in stealth and precision-guided weapons, the current age will not afford the United States an advantage in the technologies that are now transforming warfare: drones and artificial intelligence.

The conflict with [Iran](#) was the United States' first taste of a new era of warfare. Emerging technologies are leveling the playing field between Washington and its adversaries. The diffusion of affordable drone technology and artificial intelligence capabilities is allowing smaller states and nonstate actors the chance to punch above their weight. Such adversaries can now hit U.S. rear bases, inflicting casualties and damaging expensive U.S. aircraft. Iranian missile attacks on U.S. bases in the Gulf destroyed one E-3 Sentry early warning aircraft. That loss is even greater than the airplane's \$300 million cost, since the U.S. fleet of E-3 aircraft is now down to only 15 and a replacement program is years away. Iranian missiles struck five KC-135 Stratotanker refueling aircraft, as well as multiple U.S. ground radars.

Drones have transformed not just the dynamics of warfare but also its economics. In the Gulf and elsewhere, low-cost air and naval drones and missiles can take out far more expensive assets. [Ukraine](#) used kamikaze drone boats and antiship missiles to eviscerate Russia's Black Sea Fleet, sinking 13 ships after two years of war and damaging dozens more. A \$300,000 drone boat can cripple a navy warship that costs hundreds of millions of dollars.

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The United States still has the most powerful military in the world, but it is not yet prepared for a new age of warfare defined by these realities. It needs to produce more low-cost drones and interceptors, and it needs to better adapt to the imperatives of AI competition. Just as the military can't amass airpower without building planes and can't dominate the seas without floating ships, it can't win in the AI age without harnessing data, buying computing power, and learning how best to use AI models. To maintain an edge on the battlefield, the U.S. military must find ways to efficiently assimilate these new technologies. That will require overcoming cultural and bureaucratic barriers within the armed services, forging closer relationships with the private sector, and finding new ways to assess military power. But if the U.S. military does not adapt in this way, it will increasingly find itself closely matched on the battlefield. After decades of dominance assured by its technological edge, the United States will be diminished because it has let its lead perilously slip.

GAME OF DRONES

The United States has long relied on technological innovation to gain an advantage over its adversaries. Early in the [Cold War](#), U.S. defense planners counted on nuclear weapons to offset the Soviet military's superior numbers in Europe. In the 1970s, the United States ushered in the information revolution in its military

planning, and advances in semiconductors, computer networks, and satellites gave it a lead in stealth systems, precision-guided weapons, and GPS. These technologies proved invaluable in the 1990–91 Gulf War, when the United States systematically dismantled the Iraqi military. Their effect was even more impressive during the 2003 Iraq invasion, when U.S. forces seized Baghdad in just three weeks. In 2014, the Pentagon launched the “third offset” strategy, which sought to use robotics and AI to make up for the numerical superiority of Chinese and Russian forces. This strategy pushed the U.S. military to harness AI technology emerging from the commercial sector and persuaded U.S. officials that they could cement an enduring technological advantage over adversaries.

But this time, such a strategy will not work. The United States no longer has a discernible advantage in emerging technologies and will not be able to gain one.

Take, for instance, uncrewed vehicles. Cheap drones are widely available around the world, and the United States will not be able to prevent competitors from fielding them in large numbers. Iran has emerged in recent years as a major producer of cheap drones and has supplied thousands of drones to Russia for its [war in Ukraine](#). Based on Iranian designs, Russia has produced tens of thousands more.

In theory, the United States should be able to produce a huge number of these weapons. Low-cost drones don’t rely on any special technology. But in practice, the U.S. military has struggled to field cheap drones in any significant numbers. Ukraine produces four million drones every year, while the U.S. Army is acquiring only 50,000.

Pentagon leaders in both the Biden and the Trump administrations have made the production of low-cost drones a priority, but structural problems have gotten in the way. Small military drones rely on technology originally developed for the commercial hobbyist market, which is dominated by the Chinese company DJI. The U.S. military rightly does not want to depend on military hardware from its chief competitor, so it ends up buying far more expensive U.S.-made drones (which still often use Chinese components).

Compute is analogous to manufacturing capacity in the industrial age.

More damning, the United States simply isn’t good at building anything cheaply, responding quickly, or scaling up rapidly. For decades, U.S. defense production has marched steadily up a cost curve toward ever more “exquisite” defense platforms—military parlance for advanced, expensive, and low-volume weapons. Drones, by contrast, have tilted the military landscape toward low-cost, attritable (or expendable) weapons that can be produced in great quantity.

The United States has been slow to adapt. The Defense Department’s 2023 Replicator initiative aimed to field thousands of low-cost autonomous systems quickly but yielded only hundreds. The current Pentagon leadership has announced plans to expand the production of low-cost drones, committing over \$1 billion to produce 340,000 drones by 2027. The army has set an even more ambitious goal of producing at least a million drones by 2028. To achieve these goals, the military will need to deliver consistent and substantial funding to build an industrial base for small drones that does not yet exist at significant scale.

But drone technology is not standing still. Soon, these vehicles will be able to operate with greater autonomy and in closer coordination with other machines. Most drones today are remotely piloted or use simple automation, such as by following designated waypoints or returning to base if they lose the connection to a human pilot. Ukraine has become a testing ground for more sophisticated autonomous features. For instance, many Ukrainian drones have autonomous terminal guidance, allowing the uncrewed aircraft to navigate several hundred meters on its own to the target if enemy jamming breaks the communications link between the machine and the human pilot. Ukraine is also producing long-range strike drones that can travel up to 600 miles and autonomously navigate without GPS by matching images from onboard cameras to preloaded satellite imagery. These innovations will be adopted far beyond Ukraine. More countries and nonstate actors will soon possess similar drones that can hit targets even when adversaries can block communications and prevent the drone from accessing GPS. Drones will be equipped with ever more

sophisticated autonomous guidance systems that will allow them to search wide areas and identify and attack targets all on their own.

These advances will change warfare in profound ways. What today are simple drones will become tomorrow's intelligent swarms: thousands of drones reacting in real time to changing conditions on the battlefield. Swarms will be used to hunt mobile targets, conduct simultaneous attacks to overwhelm defenses, and build communications and logistics networks that are resilient to enemy jamming, disruption, or attacks. Autonomous robot swarms will be able to act with a speed, coordination, and dynamism that human pilots could never replicate.

Taking full advantage of drone swarms will require radically rethinking military command and control, organizational structures, and how human commanders direct military forces on the battlefield. Military operators won't pilot drones directly. They will command entire swarms of hundreds or thousands of drones, with the drones themselves autonomously coordinating their behavior. Militaries will need to figure out what types of directions to give swarms and how autonomous drones should coordinate among themselves. That will require quite a change from traditional models of command in the military, replacing hierarchical structures with more decentralized ones.



A military robot, San Francisco, February 2026
ALEKSANDRA MICHALSKA / REUTERS

Drones are already changing dynamics on the battlefield in ways that the United States has not yet grappled with. In the war in Ukraine, for instance, persistent drones overhead have made it hard for either side to mass forces. Drones are now responsible for the majority of Russian casualties, supplanting artillery. The war in Iran has shown how drones have made bases far from the frontlines vulnerable. The U.S. military will have to adapt to this new reality, investing more heavily in camouflage, decoys, and other methods of hiding from detection and dispersing forces to reduce risk.

The United States also needs more cost-effective ways to defend against the vast numbers of missiles and cheap drones that adversaries can launch. Missile defense has come a long way in the 35 years since the Gulf War, when U.S. Patriot batteries were almost entirely ineffective in shooting down Iraqi Scud missiles targeting Israel. But offensive missile technology has evolved, too, and the threat from drones has

mushroomed. The net effect has been that the United States has lost ground despite running faster. Missile defenses today are effective but costly. The United States, Israel, and the Gulf countries shot down 1,700 Iranian ballistic missiles and drones since the end of February, but the cost-exchange ratio has heavily favored Iran. Intercepting a \$35,000 (or, according to some recent estimates, \$7,000) Shahed drone with a \$4 million Patriot missile will only ever be a Pyrrhic victory. Washington sees the losses mounting on the balance sheet.

The American military doesn't have enough missile interceptors, and the war against Iran has badly depleted U.S. stockpiles. Just since the war began, the United States has used roughly half of its Patriot missiles and between 50 and 80 percent of its THAAD interceptor missiles. The Trump administration is taking steps to expand production capacity, but it will take years to replenish the losses. The depletion of these stockpiles will leave U.S. forces vulnerable not only in the Middle East but also in Asia and Europe.

As with low-cost drones, the Pentagon is taking steps to develop and scale production of low-cost interceptors. U.S. Coyote drone interceptors cost around \$125,000 apiece, while Merops drone interceptors cost around \$15,000 each, a major improvement over million-dollar missiles. Washington will need to scale up production of these cheaper interceptors just to keep pace with the growing threat.

NEXT TOP MODEL

AI will bring even more sweeping changes to warfare. Although the United States is home to the world's leading AI companies, advances in the field will further accelerate the erosion of American military technological superiority. Washington is gripped by the supposed "AI race" between the United States and China, but the reality today is essentially technological parity.

Chinese AI models trail leading American ones by only a few months. Chinese companies such as DeepSeek, Moonshot, and MiniMax effectively piggyback on U.S. models, using them to train their own models at a fraction of the cost. Anthropic, OpenAI, and Google have all caught and reported foreign competitors that were conducting large-scale efforts to extract information from American models in violation of those models' terms of service. Chinese companies make up for their limited access to advanced AI chips—constrained by U.S. export controls—by copying the gains made by U.S. firms that possess the most powerful and advanced chips. This technique, called adversarial distillation, functionally negates the American advantage in the most cutting-edge AI capabilities.

Another area in which the United States has until recently enjoyed an edge is in using AI to transform its intelligence analysis and operational planning. Large language models are integrated into Palantir's Maven Smart System, which pools intelligence from multiple sources into a single interface for analysts to assess the battle space. AI allows intelligence analysts and planners to synthesize vast amounts of data and plan strikes. The Israeli military reportedly used machine learning systems to process data and recommend targets for strikes in Gaza, but the U.S. military's operations against Iran are likely the first significant use of large language models on the battlefield. In Iran, where U.S. warplanes have frequently been redirected to new targets midflight, the U.S. military has used AI to prioritize targets and build strike packages amid a fluid and dynamic battle space.

America's technological dominance is waning.

But within a few months, China's military will have access to AI models with the same capabilities. In fact, every military and nonstate group on the planet will have access to these kinds of tools; after all, AI is not the closely guarded secret of particular governments, but the work of the commercial sector, and such innovations proliferate worldwide fairly quickly. Even though leading American companies are willing to work with the U.S. military, AI technology spreads faster than the military can reasonably integrate and adopt it, never mind use it to transform operations. Indeed, what matters more for militaries is not which country first develops a new AI tool or capability, but which military can first adopt it.

During periods of disruptive technological change, what determines a military's relative success is how well it employs new technology. In the early twentieth century, for instance, all the leading military powers of the

age had access to new weapons such as tanks, submarines, and airplanes. The challenge was figuring out how best to use them.

The period between World War I and World War II saw militaries experiment with new technologies and invent new organizational structures, doctrines, and training to harness these weapons. The United Kingdom was the first to innovate with aircraft carriers but fell behind Japan and the United States in the run-up to World War II. British aircraft technology was among the most advanced, but cultural and bureaucratic obstacles within the British military, such as its misguided decision to give responsibility for naval aviation to the Royal Air Force rather than the navy, slowed technological adoption.

That matters because methods, more than cutting-edge equipment and systems, make the difference on the battlefield. After all, most wars are fought between adversaries that have approximate technological parity. In a study of land wars from 1956 to 1992, the scholar [Stephen Biddle](#) found that the time gap between adversaries in military technology was on average less than three years.

AT THE BLEEDING EDGE

Restraining China's computing power is essential to edging out Beijing in AI adoption and allowing the U.S. military to use AI more effectively, even if China has access to AI models with the same capabilities. Computing power is essential for deploying AI at scale. Using the most advanced AI models takes a lot of energy and computing power, and tech companies are pouring hundreds of billions of dollars into building massive data centers to meet AI demand. Today, computing power is roughly analogous to manufacturing capacity during the industrial age. Just as a country's manufacturing capacity determined its economic growth and military prowess, aggregate "compute" will determine a country's AI power—and, consequentially, its strength.

The most powerful tool the United States has to slow China's progress in AI is export controls that prevent Chinese firms from procuring advanced chips and semiconductor manufacturing equipment. Chips are essential for training and using the most advanced AI models, and U.S. companies occupy key chokepoints in the chip production supply chain.

Under the first Trump administration and the Biden administration, the U.S. government steadily ratcheted up export controls on advanced AI chips and chip-making equipment to China. But in January 2026, the Trump administration reversed course and approved Nvidia's H200 chip for sale to China. As of April 2026, the chips had not yet been transferred to China, even though the Commerce Department had issued licenses for limited quantities and Nvidia had received orders from Chinese customers. Given overall constraints in the supply of chips for AI development and surging demand in the United States, every chip sold to China represents a loss for Washington and a boon for Beijing. The Trump administration should reinstate the ban on advanced AI chips to China rather than give up the United States' lead to a strategic competitor.

The Trump administration should also work with Japan and the Netherlands to tighten export controls on chip-making equipment to China. Advanced chip fabrication plants rely on technology from Japan, the Netherlands, and the United States. China is desperately trying to increase its domestic semiconductor manufacturing capacity to reduce its dependence on foreign chips. But without access to critical chip-making equipment, China will not be able to produce leading-edge chips. The first Trump administration put significant pressure on the Netherlands to halt sales of extreme ultraviolet lithography equipment to China, machines that are needed to make the most advanced chips. China has nevertheless continued to make progress by using older, deep ultraviolet immersion lithography technology that is not restricted.



A soldier piloting a drone, Hohenfels, Germany, April 2026
ANGELIKA WARMUTH / REUTERS

Of course, trying to restrict China's access to hardware, such as chips and chip-making equipment, will do little to limit its gains from adversarial distillation. The U.S. government should also work with AI companies to crack down on foreign competitors that extract the capabilities of American models. Congress should pass legislation to protect U.S. companies from antitrust liability when they share information about adversarial distillation with one another, similar to existing legislation addressing cyberthreats. Better cooperation among American AI firms could improve defenses against adversarial distillation by sharing best practices and threat information. And Washington should sanction Chinese entities involved in illicitly extracting the capabilities of AI models belonging to U.S. firms. Sanctioning specific Chinese companies would prohibit U.S. firms from working with them and, in the most extreme case, cut the offending Chinese companies off from the global financial system.

In some cases, AI labs themselves may want to keep some of the most advanced AI capabilities from public release, which could slow proliferation. OpenAI and Anthropic have taken this approach when delaying the release of their latest models, such as Anthropic's Mythos, out of fear that bad actors could use them for offensive cyberattacks. Anthropic has partnered with several leading technology companies in Project Glasswing to use Anthropic's AI model to find and patch cyber-vulnerabilities before more dangerous capabilities proliferate. OpenAI has created a "trusted access" program that allows thousands of verified cybersecurity experts to access OpenAI's tools for cyberdefense.

These approaches can give cybersecurity professionals a head start in fending off the dangerous AI capabilities that are coming, but the clock is ticking. As of October 2025, the AI research group Epoch AI assessed that the most capable open-weight models—that is, models available for anyone to download—trailed state-of-the-art models by only three months. Restricting release will slow proliferation by making adversarial distillation more challenging, but it will not be a permanent solution. Jack Clark, the co-founder of Anthropic, estimated in April 2026 that what counts as state-of-the-art AI cyber-capabilities today will be broadly available and openly sourced within 12 to 18 months.

Methods, more than equipment, make the difference on the battlefield.

Washington cannot halt the proliferation of AI capabilities, but it can still gain a little bit of an edge. Stretching a three-month lead into 18 months buys more time for cybersecurity experts and the U.S. military to adopt the latest AI technologies. In that sense, the right approach to technology won't give the United States an enduring advantage, but it will offer Washington a small lead in what will be a constant race.

The United States needs to use that time to innovate, experiment with AI, and adapt its own organizations and doctrine to make the most of the latest technology. Doing so will require a mindset shift, steering away from the ponderous, deliberate approach the U.S. military usually takes in peacetime to a wartime approach based on swift iteration and adaptation. The U.S. military rapidly revised its practices during the wars in Iraq and Afghanistan, quickly fielding equipment and modifying tactics to counter the threat from improvised explosive devices and to fly drones to surveil insurgents. Traditional bureaucratic Pentagon processes for establishing requirements for military systems, budgeting costs, and procuring technologies won't keep pace with AI and stay ahead of adversaries. Motivated by an existential sense of urgency, Ukraine has scaled production to four million drones a year. With 140 times Ukraine's GDP, the United States should be able to come close to that number. Although it took years for the Pentagon to invest sufficiently in armored vehicles to seriously counter the threat from roadside bombs in Iraq and Afghanistan, once Secretary of Defense Robert Gates made it a priority in 2007, the military fielded 10,000 armored vehicles in about a year and a half.

Fortunately, the current leadership in the Pentagon is willing to break the mold. The Department of Defense has put large language models on its classified and unclassified networks, giving three million military and civilian users across the defense establishment access to AI models. Pentagon leadership is also expanding the number of models available across networks, giving employees access to a diversity of AI platforms. Initial signs are positive. The Defense Department has reported that over one million users have used AI models. But the department will have to do more to create the right bureaucratic and cultural incentives for adoption. This includes giving employees the freedom to experiment with AI and accepting failure and mistakes.

The department's AI strategy, released in January, emphasized the importance of speed. To help cut through red tape, the strategy established a monthly "barrier removal board" to waive nonlegislative restrictions that might impede AI adoption. To allow greater access to data, the strategy directed that data be shared with authorized users and that any denial of a request for data be justified within seven days. These are welcome moves to speed up the Pentagon. But speed alone won't be enough.

IDENTITY CRISIS

Some of the biggest obstacles to fully harnessing the advantages of new technology are cultural. Technological advances require new ways of waging war, and these can sometimes challenge ingrained habits and deeply held identities within the military services. The U.S. Navy resisted the transition from sail to steam in the nineteenth century and even regressed on steam adoption after the Civil War. Debates about how to most effectively use tanks persisted in the U.S. Army throughout World War II. As late as 1943, Lieutenant General Lesley McNair, the commander of Army Ground Forces, wrote a memo to General George Marshall, the army chief of staff, arguing that Germany's blitzkrieg through France three years earlier was an aberration, and that the proper role of tanks was to support the infantry, not lead an armored assault on their own.

Today's military services are no less hidebound. Each service's culture and conception of airpower shape how it has adopted drones. The army was the first to embrace more automated flight controls, including for takeoff and landing, and to use enlisted personnel as drone controllers. The air force resisted these innovations, which challenged its conception of drone controllers as "pilots." Yet the air force was innovative in piloting drones from bases in the continental United States while the army chose to forward deploy drone operators to Iraq and Afghanistan, a much less efficient use of personnel. Concentrating drone operators on bases in the United States allows them to operate drones continuously, while the army's policy of forward-deploying drone operators during the wars in Iraq and Afghanistan meant roughly two-thirds of army drone

operators were stateside between deployments and not flying. But in the army's view, soldiers shouldn't telecommute to war.

Enthusiasm for uncrewed and robotic systems has varied widely in the navy. The navy's submarine force has largely embraced undersea robotic vehicles, which are a complement to submarines, not a substitute for them. In naval aviation, however, aircraft carrier deck space is limited. Each drone added to a carrier deck supplants a traditional crewed fighter aircraft. Even though a stealthy combat drone could dramatically extend the carrier's reach, the navy downgraded its carrier-based drones to tanker aircraft that would transport gas to support, not replace, crewed fighter aircraft. In doing so to save pilots' jobs, the navy chose to sacrifice the aircraft carrier's reach and striking power.

Artificial intelligence presents an even greater challenge to the self-images of military services than do drones. AI raises fundamental questions about the roles of humans and machines. The same fears about AI taking jobs across society will play out in the military, where service members' identities are strongly connected to the tasks they perform—so strongly that they sometimes persist even after technology has long rendered a task obsolete. Naval personnel are still called “sailors” even though they no longer climb masts, lower or raise sails, or handle rigging. The army still has soldiers who identify as “cavalry” even though they no longer ride horses. These identities persist as historical artifacts even as the jobs of military personnel change—and the same could happen as AI transforms the armed services. But the history of military technological adoption, from steam-powered ships to tanks to drones, suggests that identity and culture can be powerful forces preventing militaries from unlocking the true benefits of new technologies.

SINKING THE ARMADA

Another force in the United States is essential to ensuring the country's military technological lead: the private sector. Adopting AI that works will require deep partnership with the broader industry, the companies developing AI, and third-party evaluators who are experts in AI capabilities and limitations. To do so, Pentagon leadership will need to repair relationships with Silicon Valley that have grown strained in recent months by the falling out with Anthropic over the terms of its contract with the Defense Department—the Pentagon insisted it wanted unrestricted access to Anthropic's technology for “any lawful use,” while Anthropic wanted to put guardrails around the potential use of its technology for domestic mass surveillance and for powering fully autonomous weapons. At stake is much more than just the military's ties to one company. The public dispute has fed a backlash among AI engineers, who are now increasingly opposed to working with the military. Over 1,000 employees at Google and OpenAI signed an open letter urging their companies to “stand together to continue to refuse the Department of War's current demands.” In April 2026, more than 600 Google employees signed an open letter urging the company not to allow its AI models to be used for any classified work at all. Senior defense leaders have mismanaged this crisis and reignited long-standing tensions between the military and the AI industry.

The Defense Department cannot afford to alienate the engineers who are building the most powerful technology that will shape the future of war. The military must have access to leading-edge AI, but coercing U.S. companies, as the Pentagon tried to do by labeling Anthropic a “supply chain risk,” won't help encourage collaboration. After Google discontinued work on the Defense Department's early machine-learning and data-integration initiative known as Project Maven in 2018, the Pentagon went on a charm offensive. It produced the AI Ethical Principles, the department's guidelines for responsibly adopting AI, that not only helped address many AI researchers' concerns about the military applications of their work but also improved the military's processes for using AI. Today's Pentagon leadership must urgently change course to defuse tensions and build bridges, not burn them.

AI is powerful but has many flaws. Large language models today have subtle biases, tend to make things up, and engage in sycophancy—telling the user what the AI believes the user wants to hear. The effective use of AI requires grappling seriously with these limitations. AI agents, which can take independent actions on computers and networks, will accelerate productivity. They can also go badly awry. In April 2026, an AI agent deleted one company's entire database in nine seconds. (The AI agent had the grace to apologize afterward.) The military will need to set guardrails for AI systems and agents, as well as provide training for human users to ensure that AI does not lead to damaging mistakes. The military needs not just to win over

AI researchers but to actually listen to them to better understand the technology's limitations. Partnership with industry is essential to establishing the benchmarks, standards, and testing processes needed to make the military's use of AI successful.

Washington cannot halt the proliferation of AI.

Finally, the armed forces must update its metrics for measuring military power in this new era. The navy counts the number of ships; the air force, the number of aircraft. These are industrial-age metrics. (The army counts the number of soldiers—a pre-industrial metric.) For one thing, planners must do a better job of factoring low-cost drones into these counts. Often, these vehicles are not considered powerful enough to count as aircraft, but excluding them risks understating military capacity and skewing planning toward legacy systems.

But far more important than these figures now are measures of the digital components that empower and connect military platforms: sensors, radars, computers, networks, and algorithms. The Defense Department should begin tracking AI-relevant metrics. These could include the amount of computing power it has available at any one time on classified and unclassified networks and how much that computing power is being used. It could also track active monthly users, token usage on AI models to show how widespread and frequent AI use is, and the amount of data available across the Defense Department and how it is being used. These figures would give planners a more detailed understanding of the extent to which military and civilian workers are using AI and where additional investments or initiatives are needed to accelerate adoption. Just as the number of ships, aircraft carriers, planes, and service members are points of discussion in the Defense Department budget, so, too, should the number of H100-equivalent GPUs the department has access to. To lead in AI, the military will need to invest in AI power. The military should also conduct detailed assessments of the use of AI to measure whether the technology has increased efficiency and accuracy, improved costs, and accelerated workflows, as well as to determine what lessons can be applied to other applications.

History is full of cautionary tales of militaries that struggled to shift and reform after the advent of disruptive technologies. When English and Spanish fleets clashed in 1588, Spain was at the height of its power. But the English navy had more successfully capitalized on the new technology of the day: cannons. The Spanish Armada, by contrast, was still designed around the imperative of closing with and boarding enemy ships, their decks packed with infantry. As a result, the vast Spanish fleet was hopelessly outgunned and was defeated. The war between England and Spain dragged on for 16 more years after the defeat of the Spanish Armada, but the peak of Spain's naval power had passed, as had the peak of Spain's power as a global empire.

The United States can remain the world's leading military if it acts now to adapt to the changing contours of modern warfare. But if the Pentagon fails to push its operations in the necessary directions, it will be eclipsed by competitors that are more dogged and intrepid in adjusting to the realities of a new age.